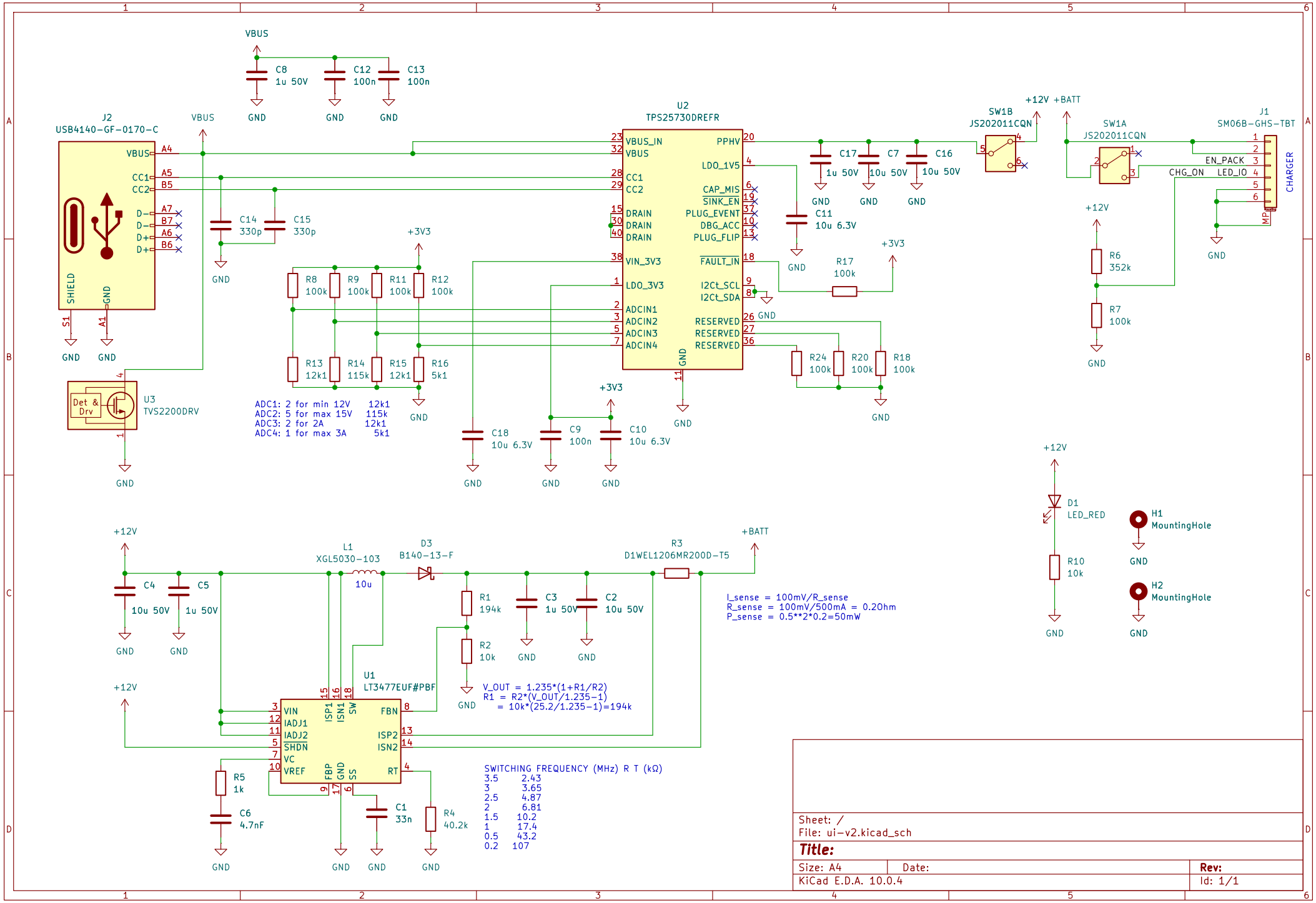

VBUS
C8
1u 50V
C12
100n
C13
100n
GND
J2
USB4140-GF-0170-C
VBUS
CC1
CC2
D-
D-
D+
D+
SHIELD
GND
S1
A1
C14
330p
C15
330p
U2
TPS25730DREFR
VBUS_IN
VBUS
CC1
CC2
DRAIN
DRAIN
DRAIN
VIN_3V3
LDO_3V3
ADCIN1
ADCIN2
ADCIN3
ADCIN4
PPHV
LDO_1V5
CAP_MIS
SINK_EN
PLUG_EVENT
DBG_ACC
PLUG_FLIP
FAULT_IN
I2Ct_SCL
I2Ct_SDA
RESERVED
RESERVED
RESERVED
GND
+3V3
R8
100k
R9
100k
R11
100k
R12
100k
R13
12k1
R14
115k
R15
12k1
R16
5k1
ADC1: 2 for min 12V 12k1
ADC2: 5 for max 15V 115k
ADC3: 2 for 2A 12k1
ADC4: 1 for max 3A 5k1
C18
10u 6.3V
C9
100n
C10
10u 6.3V
C11
10u 6.3V
C17
1u 50V
C7
10u 50V
C16
10u 50V
+12V
+BATT
SW1B
JS202011CQN
SW1A
JS202011CQN
J1
SM06B-GHS-TBT
EN_PACK
CHG_ON
LED_IO
CHARGER
R17
100k
R6
352k
R7
100k
R24
100k
R20
100k
R18
100k
U3
TVS2200DRV
Det &
Drv
L1
XGL5030-103
10u
D3
B140-13-F
R3
D1WEL1206MR200D-T5
C4
10u 50V
C5
1u 50V
R1
194k
R2
10k
C3
1u 50V
C2
10u 50V
I_sense = 100mV/R_sense
R_sense = 100mV/500mA = 0.20hm
P_sense = 0.5**2*0.2=50mW
U1
LT3477EUF#PBF
VIN
IADJ1
IADJ2
SHDN
VC
VREF
ISP1
ISN1
SW
FBN
ISP2
ISN2
FBP
GND
SS
RT
V_OUT = 1.235*(1+R1/R2)
R1 = R2*(V_OUT/1.235-1)
= 10k*(25.2/1.235-1)=194k
R5
1k
C6
4.7nF
C1
33n
R4
40.2k
SWITCHING FREQUENCY (MHz) R T (kΩ)
3.5 2.43
3 3.65
2.5 4.87
2 6.81
1.5 10.2
1 17.4
0.5 43.2
0.2 107
D1
LED_RED
R10
10k
H1
MountingHole
H2
MountingHole
Sheet: /
File: ui-v2.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 1/1